

FENGNING, China

WMO: 543080

Lat: 41.2018N

Lon: 116.6343E

Elev: 736

StdP: 92.79

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-18.0	-30.7	0.2	-13.6	-28.9	0.3	-12.9	12.4	-11.5	11.1	-11.3	1.8	320	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.0	18.9	30.4	18.8	29.1	18.8	23.3	27.8	22.3	26.7	21.4	25.7	3.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.9	18.1	26.0	21.0	17.1	25.2	20.0	16.1	24.2	73.8	27.9	70.0	26.8	66.3	25.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	8.9	7.5	-22.9	35.0	1.9	2.1	-24.2	36.5	-25.3	37.8	-26.3	39.0	-27.7	40.5
			-23.3	24.6	1.7	1.1	-24.6	25.4	-25.6	26.0	-26.6	26.7	-27.8	27.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-9.7	-5.9	1.5	9.7	16.0	20.5	23.1	21.6	16.0	8.3	-0.8	-8.2	
	DBStd	12.06	3.39	4.08	4.50	4.34	3.41	2.72	2.22	2.46	3.38	3.78	4.35	3.46	
	HDD10.0	2352	612	444	266	57	2	0	0	0	2	77	326	566	
	HDD18.3	4226	871	677	523	261	86	11	1	4	83	310	576	824	
	CDD10.0	1526	0	0	2	48	189	316	406	358	182	26	0	0	
	CDD18.3	358	0	0	0	1	14	77	148	104	14	0	0	0	
	CDH23.3	3352	0	0	1	27	313	796	1222	824	167	2	0	0	
	CDH26.7	1014	0	0	0	3	81	262	417	222	29	0	0	0	
Wind	WSAvg	2.7	2.7	2.8	3.2	3.4	3.3	2.5	2.2	2.0	2.1	2.5	2.4	2.6	
Precipitation	PrecAvg	463	1	3	6	16	34	73	132	121	52	18	5	1	
	PrecMax	659	14	24	23	60	81	163	269	256	114	45	27	8	
	PrecMin	290	0	0	0	0	3	25	40	59	13	0	0	0	
	PrecStd	91	3	4	6	13	21	33	55	58	27	13	7	2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	9.4	19.7	26.3	31.7	33.9	35.0	32.6	29.6	23.0	14.1	4.6	
		MCWB	-2.4	1.5	7.4	12.1	15.5	17.3	20.2	20.6	17.2	12.6	5.9	-1.3	
	2%	DB	0.7	6.7	15.7	23.5	29.0	31.5	32.3	30.5	27.1	20.0	11.2	2.1	
		MCWB	-4.0	0.0	5.7	10.9	14.6	17.2	20.3	19.9	16.7	11.4	4.2	-2.4	
	5%	DB	-1.1	4.6	12.9	21.2	26.8	29.6	30.5	29.1	25.2	17.9	9.0	0.2	
		MCWB	-5.3	-1.2	4.2	10.2	13.6	17.1	20.1	19.8	15.7	10.1	3.1	-3.9	
	10%	DB	-2.9	2.4	10.3	18.9	24.7	27.8	27.8	29.0	27.7	23.3	15.9	6.8	-1.6
		MCWB	-6.5	-2.4	2.6	8.7	12.6	16.9	20.4	19.6	15.1	9.1	1.9	-5.0	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.0	2.7	8.6	15.1	18.7	21.8	24.7	24.3	20.7	15.2	6.9	-0.4	
		MCDB	2.8	7.4	17.1	21.7	27.0	28.0	29.6	29.1	26.1	19.7	10.8	4.1	
	2%	WB	-3.9	0.4	6.4	12.9	16.7	20.4	23.7	23.0	19.0	13.2	5.1	-2.4	
		MCDB	0.5	5.7	14.7	21.1	24.6	26.1	28.3	27.5	24.2	17.6	9.9	1.7	
	5%	WB	-5.2	-0.9	4.5	11.1	15.5	19.4	22.8	22.1	17.7	11.7	3.7	-3.6	
		MCDB	-1.3	4.4	12.0	18.9	22.7	25.3	27.2	26.0	22.5	15.8	8.5	-0.1	
	10%	WB	-6.3	-2.4	2.9	9.6	14.2	18.5	22.0	21.1	16.5	10.2	2.0	-4.9	
		MCDB	-3.0	2.3	9.5	17.2	21.4	24.5	26.4	25.1	21.0	14.4	6.5	-1.6	
Mean Daily Temperature Range	5% DB	MDBR	12.1	12.9	13.2	13.9	14.2	12.5	10.5	11.2	12.7	12.5	11.8	11.7	
		MCDBR	14.6	16.6	17.7	18.1	19.0	16.8	14.4	13.9	15.7	15.5	14.7	14.1	
		MCWBR	10.8	11.2	10.3	8.7	7.7	5.8	4.9	5.3	6.5	7.9	8.8	10.2	
	5% WB	MCDBR	14.3	15.7	16.8	14.6	13.9	11.5	9.5	9.6	11.5	12.5	13.7	13.7	
		MCWBR	10.7	10.7	10.0	7.5	6.4	5.1	4.4	4.6	5.9	7.3	8.6	10.0	
Clear-Sky Solar Irradiance	taub	0.309	0.381	0.417	0.486	0.541	0.644	0.616	0.578	0.517	0.449	0.347	0.316		
	taud	2.283	2.070	2.014	1.896	1.793	1.643	1.749	1.810	1.882	1.979	2.223	2.274		
	Ebn at Noon	832	802	819	789	755	678	692	704	713	714	770	785		
	Edh at Noon	90	131	156	189	214	250	223	204	177	142	96	84		
All-Sky Solar Radiation	RadAvg	2.56	3.43	4.68	5.49	6.12	5.54	5.22	5.10	4.45	3.50	2.58	2.20		
	RadStd	0.18	0.29	0.38	0.37	0.39	0.50	0.35	0.32	0.27	0.25	0.24	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	-0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page